

Exercises

1. Explain why feedforward neural networks are not suitable for sequential data. In your answer, describe how Recurrent Neural Networks (RNNs) address this limitation.
2. Derive the forward pass equations of a vanilla RNN. Clearly define all variables involved in the computation of the hidden state and output.
3. Explain the vanishing gradient problem in RNNs. Why does it occur during backpropagation through time, and how does it affect learning long-term dependencies?
4. Compare RNNs and LSTMs with respect to memory, ability to long-term dependencies, computational complexity, and training stability.
5. Explain the role of each component in an LSTM network: Forget gate, Input gate, Cell state, and Output gate.
6. Implement the forward pass of a simple RNN from scratch using NumPy.
7. Build and train a character-level RNN using PyTorch or TensorFlow.
8. Design and implement an LSTM-based model for time series prediction.
9. Develop an LSTM-based model for binary sequence classification (e.g., sentiment analysis).
10. Design a sequence modeling system using either RNN, LSTM, or both.